Rocío Nayeli Avendaño Villeda

Marine Biologist – Ecologist – Data Analyst

CONTACT

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https://github.com/Chio360

La Paz, Baja California Sur, Mexico.

LANGUAGES

Spanish: Native 100%

English: B2 Intermediate

INTERPERSONAL SKILLS

- ✓ Teamwork skills
- ✓ Proactive
- ✓ Efficient
- ✓ Disciplined
- ✓ Punctual
- ✓ Flexible
- ✓ Responsible
- ✓ Honest
- ✓ Sociable

PROFILE

My objective is to advance both personally and professionally by integrating rigorous marine science with a deeply human, community-centered approach to ocean conservation. I seek to apply my expertise in fisheries dynamics, oceanographic data analysis, and sustainable management frameworks to protect coastal ecosystems while supporting the livelihoods that depend on them. By maximizing the value of available scientific and human resources, I aim to drive excellence in marine spatial analysis, ensuring that the interplay of physical, biological, and socio-economic processes directly informs equitable and effective marine stewardship

MARINE SPATIAL DATA ANALYST & FISHERIES ECOLOGIST

As a marine biologist specializing in fisheries bioeconomics, I contributed for two years with the fishing industry association Alianza por la Sostenibilidad de la Sardina (ALISSA A.C.) toward the Marine Stewardship Council (MSC) certification of small pelagic fisheries, specifically targeting Pacific sardine (Sardinops sagax) and Northern anchovy (Engraulis mordax). Through advanced data analysis, I provided actionable scientific evidence to evaluate stock status and ecosystem interactions, enabling industry decision-makers to adopt precautionary harvest control rules. Furthermore, I successfully trained fleet managers, vessel captains, and onboard observers on bycatch mitigation protocols, biological data collection, and MSC compliance standards. Ultimately, this analytical and capacity-building support ensured long-term resource sustainability while fully safeguarding marine ecosystem structure and dynamics.

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SKILLS

R Language: Data management, data visualization, R Markdown, classical statistics, multivariate statistics, time series analysis, image processing & remote sensing, quantitative ecology, and spatial data analysis.

Python Language: Data management, data visualization, classical statistics, and spatial data analysis/automation.

SQL: Spatial & relational database management (PostgreSQL/PostGIS) and spatial data querying.

QGIS: Geographic information visualization, cartographic map creation, and spatial dashboard maintenance/deployment (Lizmap).

Microsoft Excel: Advanced data handling, data visualization, and descriptive/classical statistics.

Adobe Illustrator: Cartographic graphic editing, map refinement, and scientific visualization formatting.

DATA ANALYSIS PROJECTS

- Processed and integrated long-term oceanographic (sea surface temperature, Chlorophyll-*a*, altimetry and upwelling) and fishery data (small pelagic landings) with seabird demographic parameters, evaluating trophic interactions and seabird bycatch to support Marine Stewardship Council-aligned ecosystem approach management.
- Spatial interactions between small pelagic fishing operations and the benthic substrate were evaluated. Spatial modeling enabled the mapping of contact zones between fishing gear and the seabed, as well as the quantification of habitat vulnerability within commercial fishing areas.
- Based on records from onboard observers, the spatiotemporal dynamics of seabird bycatch in small pelagic fishing operations were analyzed. Spatial modeling made it possible to identify specific geographic areas and seasonal peaks associated with higher interaction rates involving protected seabird species.

EDUCATION

Master´s Degree: Master´s in Science in Recourse Marine Management (2024-2026) in Centro Interdiciplinario de Ciencias Marinas, IPN.
Overall GPA: 9.77.

Thesis: “Population dynamics of the Brown Pelican (*Pelecanus occidentalis californicus*) and its relationship with biotic and abiotic factors on the islands off the West Coast of the Northern Baja California Peninsula, Mexico”.

Bachelor´s Degree: Biology (2017-2022) in Facultad de Estudios Superiores Iztacala, UNAM. Overall GPA: 8.65.

Thesis: “Changes in the functional profile of bird communities associated with the seasonality of the dry tropical forest of Alto Balsas in Guerrero”.